

Atmospheric Dynamics

- **Fundamental Forces**
- **Real forces: Pressure Gradient Force, Gravitational Force, Friction or Viscous Force;**
- **Apparent forces: Centrifugal/C.Petal Force, Coriolis force**
- **Momentum Equations; Scale analysis of momentum equations**
- **Geostrophic winds, Hydrostatic balance, Inertial Flow, Gradient wind, Thermal Wind**

Length, Velocity and Time Scales of Different Atmospheric Motions

Phenomenon	Length scale L	Velocity scale U	Time scale T
Micro turbulence	10-100 cm	5 – 50 cm/s	Few seconds
Thunderstorms	Few km	1 – 10 m/s	Few hours
Sea breeze	5 – 50 km	1 – 10 m/s	6 hours
Tornado/spouts	10 – 500 m	30 m/s	10 – 60 minutes
Cyclones	300 – 500 km	30 – 60 m/s	Days to weeks
Mountain waves	10 – 100 km	1 – 20 m/s	Days
Weather patterns	10 – 5000 km	1 – 50 m/s	Days to weeks
Prevailing winds	Global	5 – 50 m/s	Seasons to years

Laws governing the motion in the atmosphere

Atmospheric motions are governed by three fundamental laws of conservation

- ***Conservation of mass:*** Mass can neither be created nor be destroyed, although it may be rearranged in space, and changed into different types of forms.
- ***Conservation of momentum:*** Within some problem domain, the amount of momentum remains constant; momentum is neither created nor destroyed, but only changed through the action of some external forces.
- ***Conservation of energy:*** Though energy can be transformed from one form to another, it cannot be created or destroyed.

Basic Physical Principle: Newton's second law

The change in momentum of an object is equal to the sum of forces acting on that object.

$$\mathbf{F} = m \mathbf{a}$$

$$\mathbf{a} = \frac{1}{m} \sum_{\text{all } i} \mathbf{F}_i$$

$$\frac{D\mathbf{u}}{Dt} = \frac{1}{m} \sum_{\text{all } i} \mathbf{F}_i$$

What forces are important for understanding atmospheric dynamics ?

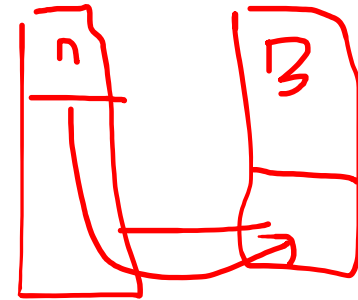
Forces that influence atmospheric motions

- **Fundamental or real forces** (those which exist regardless of our frame of reference)
 - ❖ Pressure gradient force
 - ❖ Gravitational force
 - ❖ Frictional/Viscous Force
- **Apparent forces** (those which account for the acceleration, i.e., the change in direction, of our frame of reference and which accounts for the observed motions of air/water parcels)
 - ❖ Coriolis force
 - ❖ Centrifugal force/ Centripetal force

Pressure Gradient Force

- Pressure gradient – the change in atmospheric pressure over a distance.

$$\frac{\mathbf{F}_{tot}}{m} = -\frac{1}{\rho} \nabla p$$



Total force acting on fluid parcel
in all directions

The pressure gradient results in a net force that is directed from high pressure to low pressure and this force is called the "pressure gradient force".

PG is large



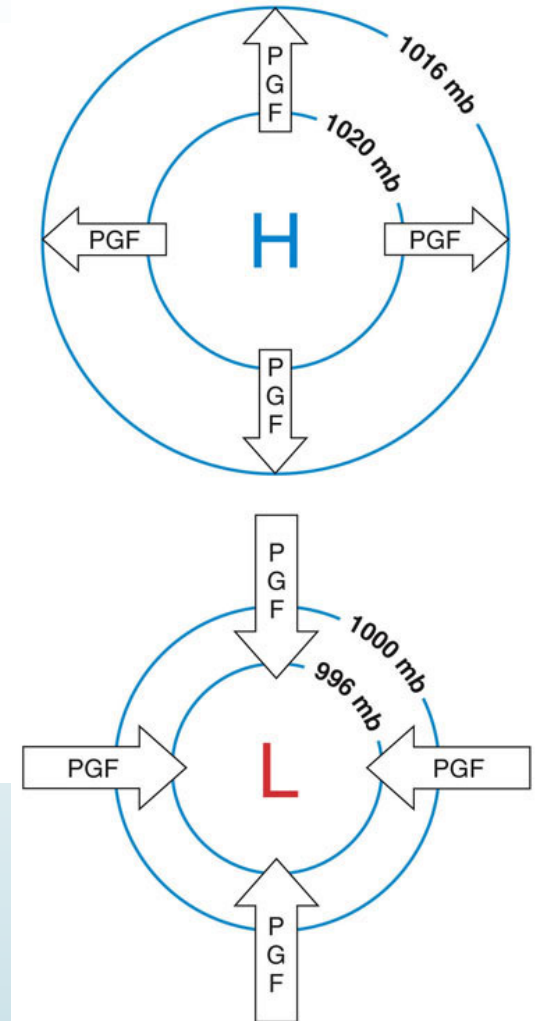
PGF is more



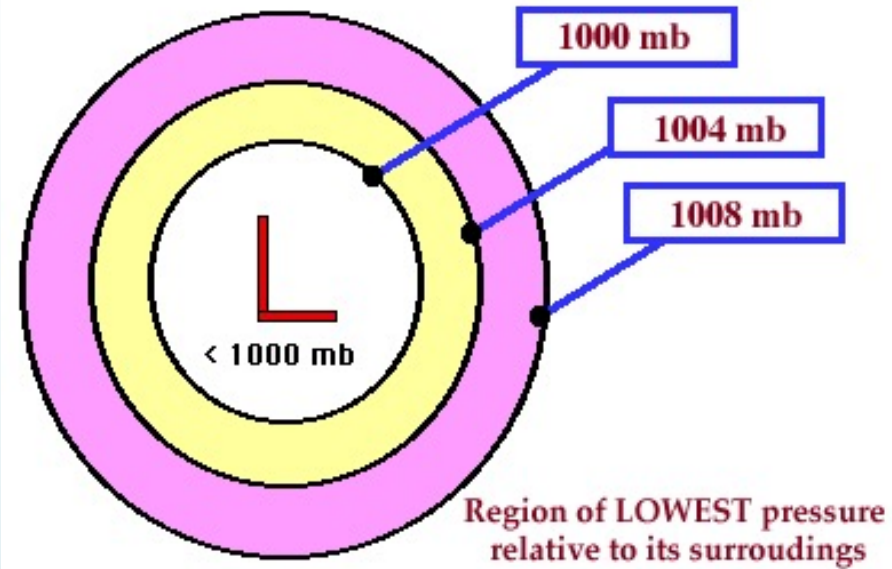
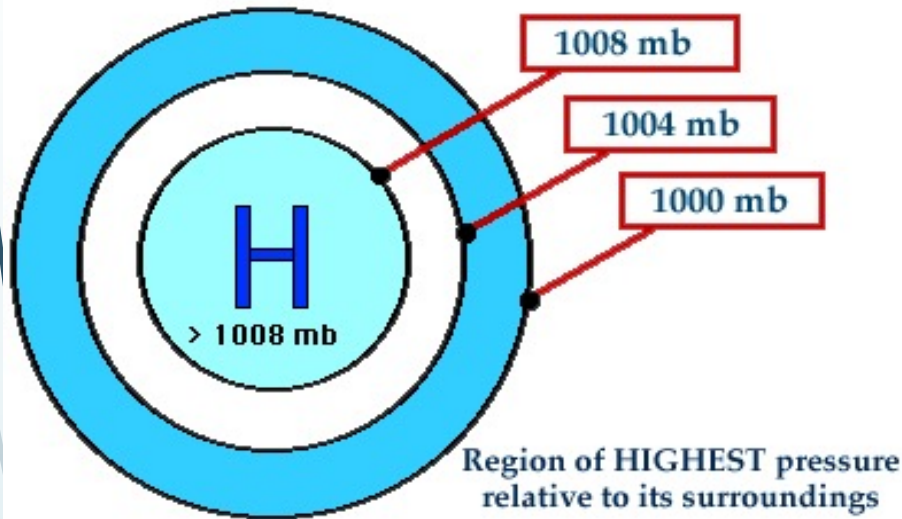
wind movement
is stronger

Pressure Gradient Force on the Weather Map

- The contour lines are called isobars, lines of constant air pressure
- H = High pressure (pressure decreases in all directions from center)
- L = Low pressure (pressure increases in all directions from center)
- Strength of resultant wind is proportional to the isobar spacing
- Less spacing = stronger pressure gradient = stronger winds

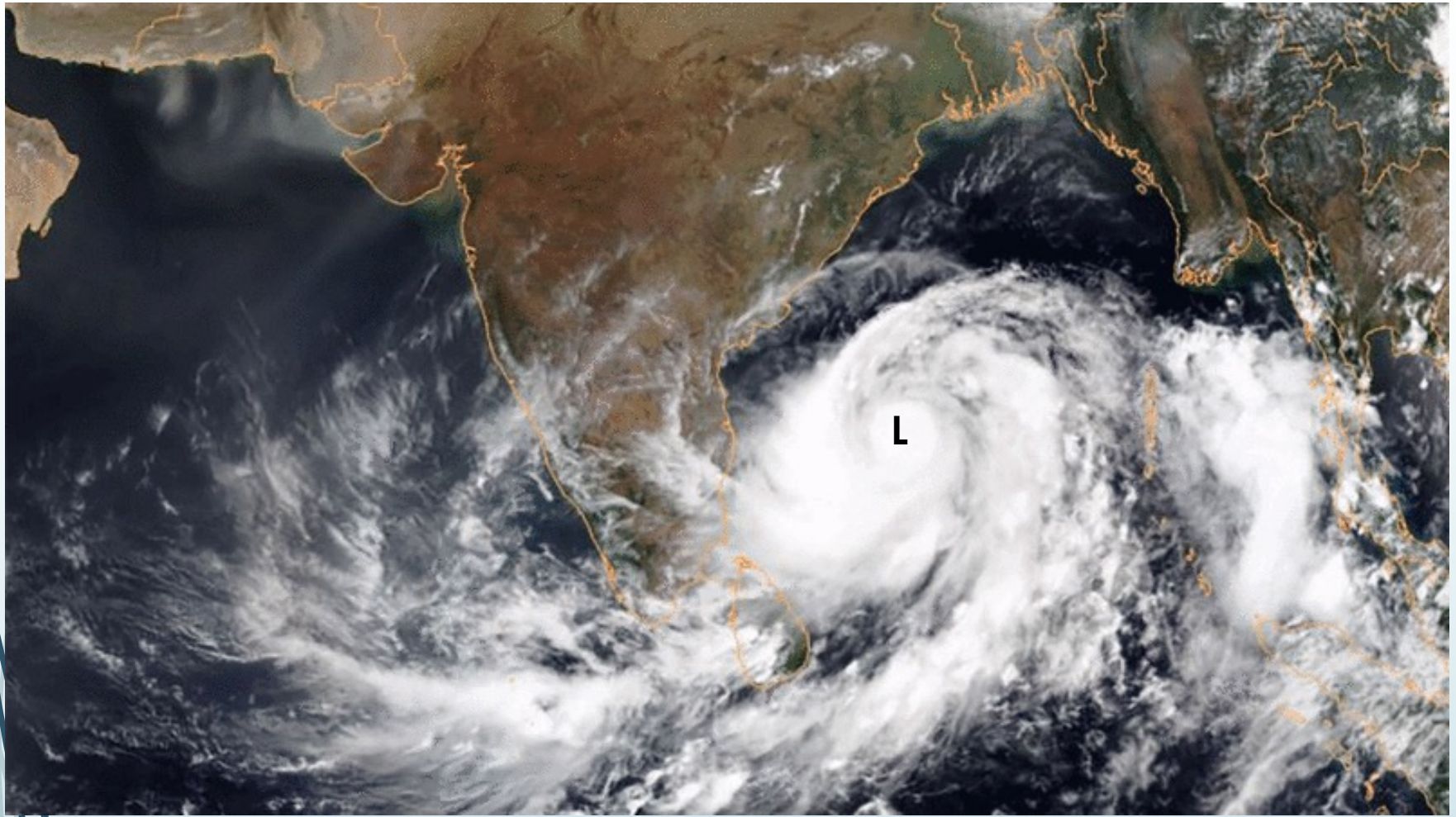


Pressure Systems: High & Low Pressure Center



- Center of highest pressure
- P decreases outward from the low center
- Clock-wise direction

- Center of lowest pressure
- Pressure increases outward from the low center
- Anti-clock wise direction



Pressure Gradient Force

$$\frac{F}{m} = -\frac{1}{\rho} \nabla P$$

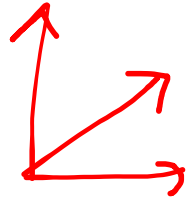
x-component of the PGF per unit mass

$$\frac{F_x}{m} = -\frac{1}{\rho} \frac{\partial p}{\partial x}$$

y- and z-components of the PGF per unit mass:

$$\frac{F_y}{m} = -\frac{1}{\rho} \frac{\partial p}{\partial y}$$

$$\frac{F_z}{m} = -\frac{1}{\rho} \frac{\partial p}{\partial z}$$

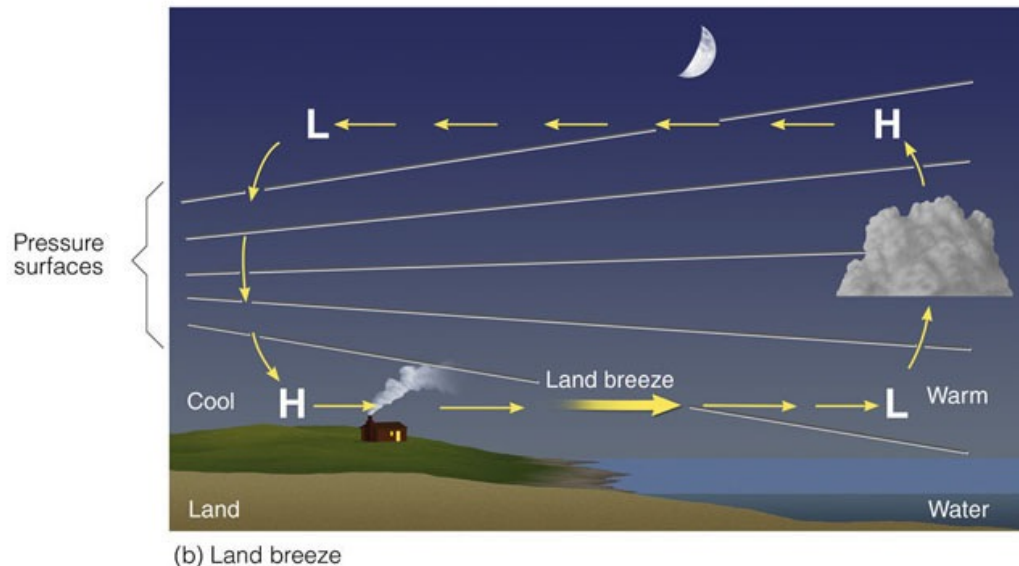
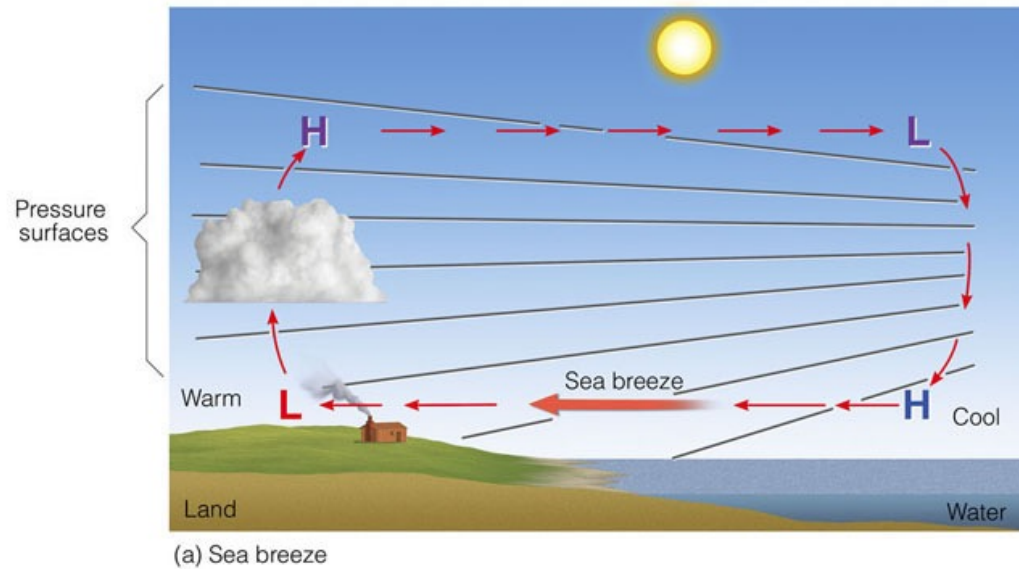


It is important to note that:

- The pressure gradient points from **low to high pressure**, but the pressure gradient force points from **high to low pressure**.
- The pressure gradient force is proportional to the gradient of the pressure field, not to the pressure itself.

Land and Sea Breezes

- Sea breeze: from sea to land
- Land breeze: land to sea
- Land and Sea Breezes also occur near the shores of large lakes



Gravitational Force

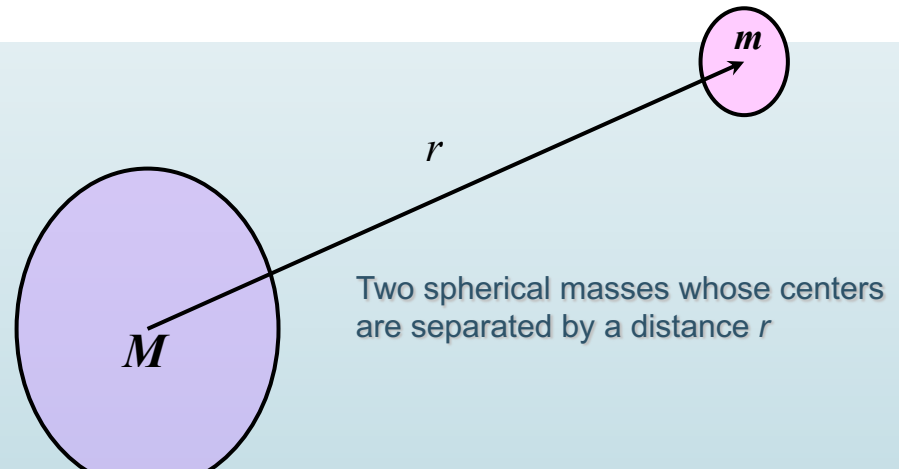
- According to Newton's law of gravitation, any two elements of mass in the universe attract each other with a force proportional to their masses and inversely proportional to the square of the distance separating them.
- If two mass elements M and m are separated by a distance $r \equiv |r|$ (with the vector r directed towards m) then the force exerted by mass M on mass m due to gravitation is: (Newton's Law of Gravity)

$$\mathbf{F}_g = - (G M_{earth} m / r^2) (\mathbf{r}/r)$$

where G is the universal gravitational constant = $6.673 \times 10^{-11} \text{ m}^3\text{kg}^{-1}\text{s}^{-2}$

The position of the air parcel relative to the centre of the Earth is $(\mathbf{r}/r)$

Mass of the earth $M=5.988 \times 10^{24} \text{ kg}$, $r=6.371 \times 10^6 \text{ m}$



- Thus, if earth is designated as mass M and m is a mass element of the atmosphere, then the force per unit mass exerted on the atmosphere by the gravitational attraction of the earth is:

$$F_g/m \equiv g = - (GM / r^2) (\mathbf{r}/r)$$

But since the atmosphere is essentially a thin shell, we can make the approximation ($r \approx a$).

Gravity at the Surface $g = GM / a^2$ ----- (A)

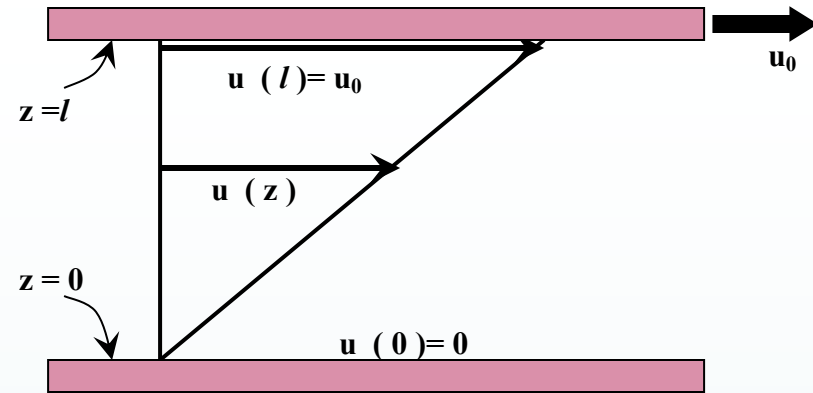
Total Gravitational force acting on the air parcel is

downward $F_g/m = -g \cdot \mathbf{k}$ -- Always constant and

Can you calculate g from the above information (A) ?

Viscous/Frictional Force

- Viscous force is the internal friction experienced by a real fluid, and which causes it to resist the tendency to flow.
- It arises due to the shearing stress resulting from a net upward/downward transport of momentum by the random motion of the molecules.



One dimensional steady –state viscous shear flow

Shearing stress: Viscous force per unit area resulting from wind shear (wind changes with Height) – viscosity is a measure of the resistance of the fluid which is deformed by shearing stress.

The force per unit mass gives the acceleration == >

For all the directions:

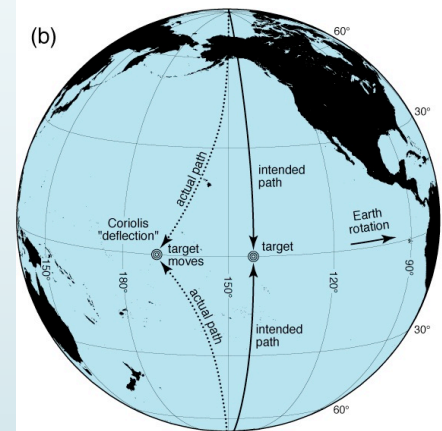
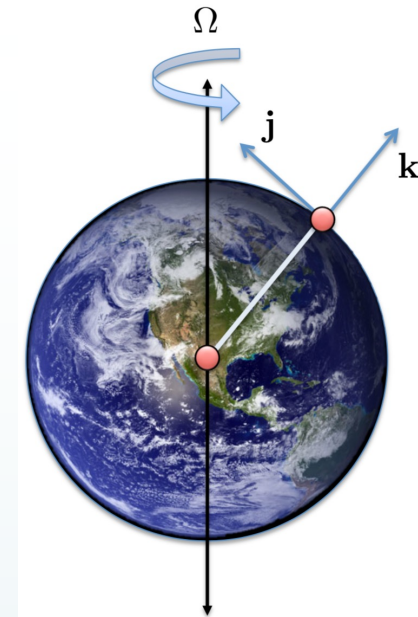
$$\begin{aligned} F_{rx} &= \nu \left[\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} \right] \\ F_{ry} &= \nu \left[\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} + \frac{\partial^2 v}{\partial z^2} \right] \\ F_{rz} &= \nu \left[\frac{\partial^2 w}{\partial x^2} + \frac{\partial^2 w}{\partial y^2} + \frac{\partial^2 w}{\partial z^2} \right] \end{aligned}$$

$$\frac{F_{tot}}{m} = \frac{\mu}{\rho} \left(\frac{\partial^2 u}{\partial y^2} \right)$$

Where μ is the dynamic viscosity

Coriolis Force

- Coriolis force arises from rotation of the earth (the conservation of angular momentum)
- If air moves in the meridional (north-south) direction, it is changing its distance to the axis of rotation (the poles).
- Let's consider this in terms of its effect on angular momentum. At a given latitude it is equal to the angular velocity times the square of the distance to the axis of rotation. Angular momentum is also conserved in rotating systems (such as the Earth)
- Thus to conserve angular momentum, the angular velocity must increase, and to increase the angular velocity, an eastward acceleration must occur.

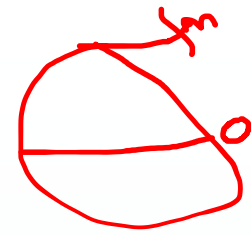


The Coriolis force only acts on an object in movement. It can't help start the movement of air, only deflect it in a certain direction once it's in movement.

The Coriolis Parameter f is defined as $f = 2 \Omega \sin \Phi$ ✓

Ω – Angular velocity

Φ – Latitude



Coriolis Force $\Rightarrow \left(\frac{D\mathbf{V}}{Dt} \right)_{Co} = -f \mathbf{k} \times \mathbf{V}$ where k is the vertical unit vector

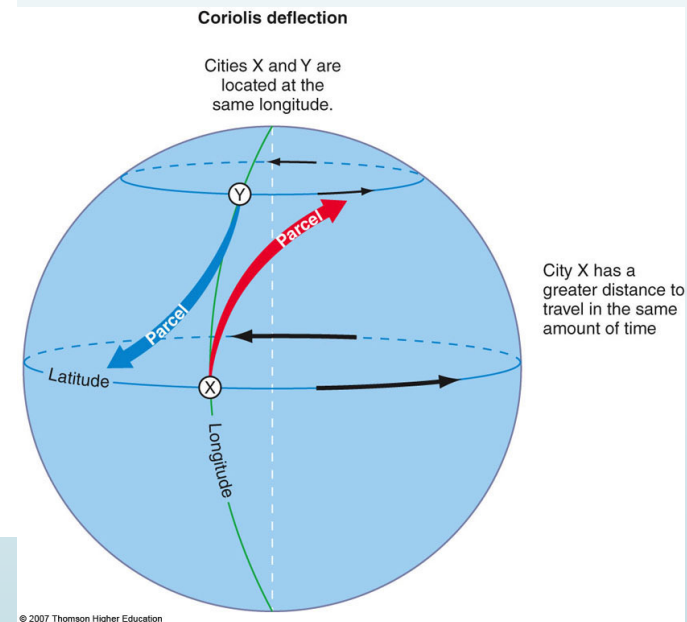
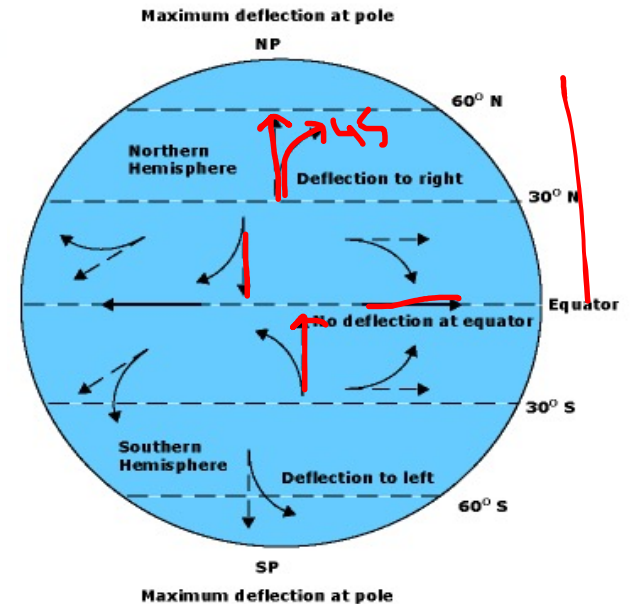
For w small, the Coriolis force is then given by:

$$\frac{Du}{Dt} = fv \quad \frac{Dv}{Dt} = -fu \quad \checkmark$$

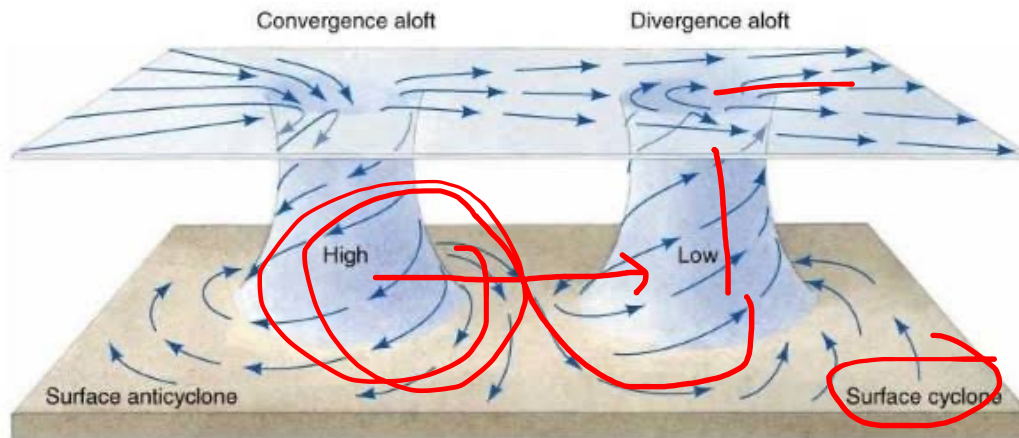
- Deflection is to the right in the Northern hemisphere and to the left in the Southern hemisphere.
- The Coriolis force can only change the direction of motion, but not the speed of motion.

Coriolis effects

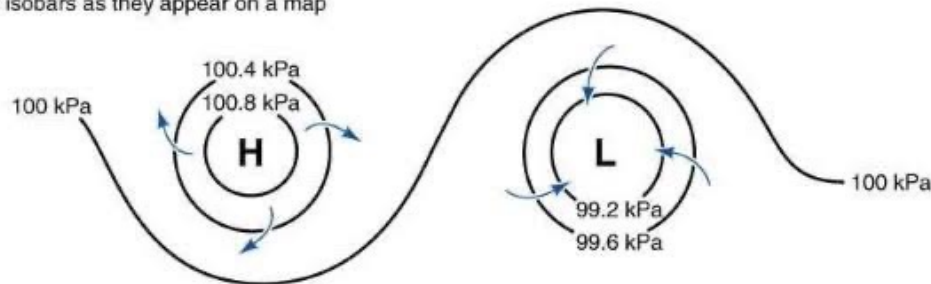
- The Coriolis force causes the wind to deflect to the right of its intended path in the Northern Hemisphere and to the left of its intended path in the Southern Hemisphere. It acts at a right angle to the wind.
- The Coriolis force is largest at the pole and zero at the equator
- The stronger the wind speed, the greater the deflection
- The Coriolis force changes only wind direction, not wind speed.
- We measure motion on the rotating Earth. Thus, we need to be concerned with the Coriolis force



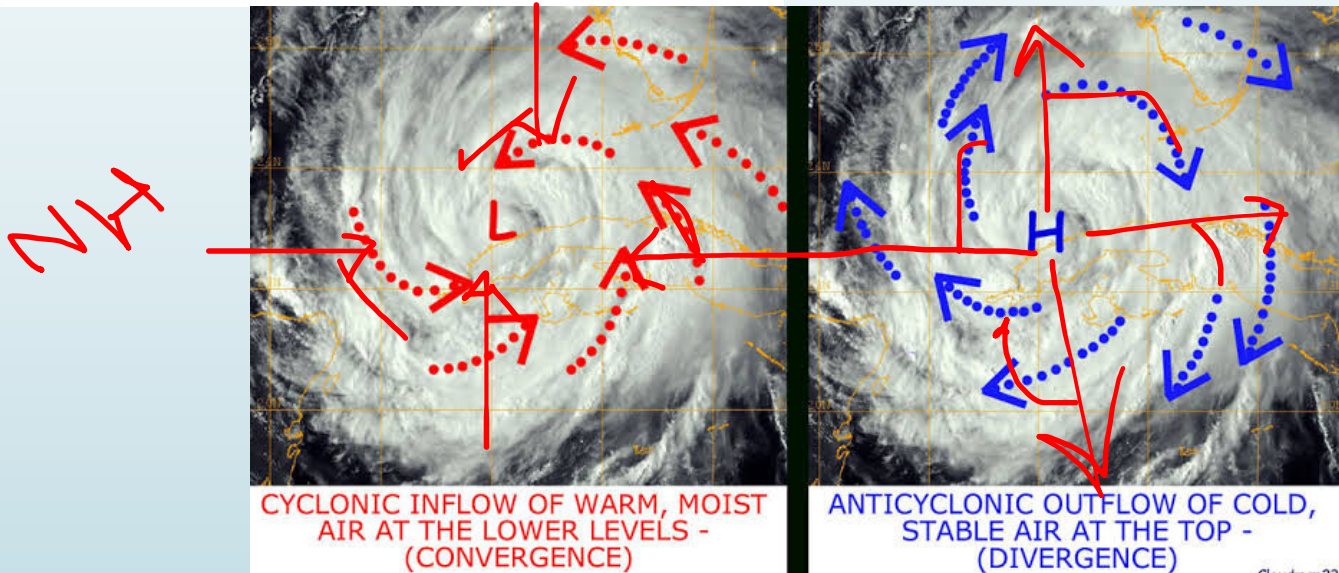
Relation with High and Low pressure centers



Surface isobars as they appear on a map

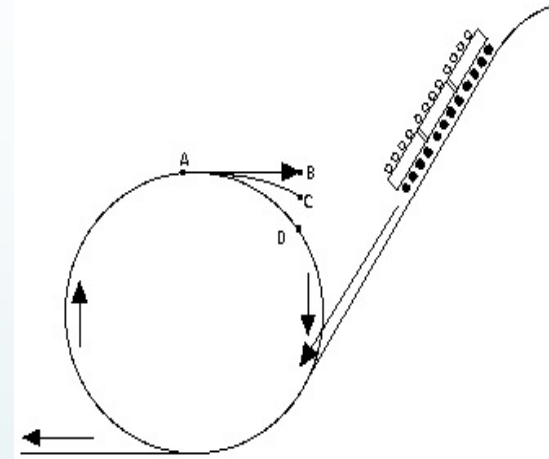


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Centrifugal Force / Centripetal Acceleration

- Due to change in direction of motion.
- A centrifugal force is a force on an object that tends to move it away from a center of rotation and always results from the **inertia** of the object.
- A centripetal force is a force on an object that tends to move it toward a center of rotation.



roller coasters in parks.

$$DV/Dt = - \omega^2 r$$

➤ Uniform acceleration which is directed towards the axis of rotation (and equal to the square of the angular velocity times the distance from the axis of rotation) viewed from fixed co-ordinates is called the centripetal acceleration.

- The centripetal acceleration is caused by the force of string pulling the ball and is applicable only if we are viewing the rotating ball from fixed co-ordinates.
- Now consider that we are observing the ball in rotating co-ordinates, that rotates along with the ball. In this system, the ball is stationary, but still there is a force that acts on the ball (the pull of the string). There will be an apparent force called centrifugal force that just balances the pull of the string.
- The centrifugal force is equivalent to the inertial reaction of the ball on the string and is equal and opposite to the centripetal acceleration.

Thus:

Observed from a fixed system, the rotating ball undergoes a uniform centripetal acceleration in response to the force exerted by the string.

And, observed from a system rotating along with it, the ball is stationary and the force exerted by the string is balanced by a centrifugal force.

Momentum Equations

Recall Newton's second law; $\frac{D\mathbf{u}}{Dt} = \frac{1}{m} \sum_{\text{all } i} \mathbf{F}_i$

Rate of change of velocity following the motion = Sum of forces acting per unit mass.

Equation of Motion

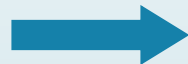
$$\frac{D\mathbf{u}}{Dt} = -2\vec{\Omega} \times \vec{U} - \frac{1}{\rho} \vec{\nabla} P + \vec{g} + \vec{F}_r$$

$$-2\vec{\Omega} \times \vec{U}$$



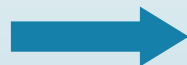
Coriolis force

$$-\frac{1}{\rho} \vec{\nabla} P$$



Pressure gradient force

$$\vec{g} = \vec{g}^* + \Omega^2 \vec{R}$$



Gravity force per unit mass = Sum of the gravitational and centrifugal force terms

$$\vec{F}_r$$



Frictional force in the fluid

Momentum Equations

Equation of Motion

$$\frac{D\mathbf{u}}{Dt} = -2\vec{\Omega} \times \vec{U} - \frac{1}{\rho} \nabla P + \vec{g} + \vec{F}_r$$

The eastward (u)_Zonal, northward (v)_Meridional and, vertical (w) components of momentum equations respectively:

Components of Equation of Motion

$$\frac{du}{dt} \quad \text{[]} = -\frac{1}{\rho} \frac{\partial P}{\partial x} + 2\Omega v \sin \phi \quad \text{[]} + F_x$$

$$\frac{dv}{dt} \quad \text{[]} = -\frac{1}{\rho} \frac{\partial P}{\partial y} - 2\Omega u \sin \phi + F_y$$

$$\frac{dw}{dt} \quad \text{[]} = -\frac{1}{\rho} \frac{\partial P}{\partial z} - g + \text{[]} + F_z$$

Not all the terms in the momentum equations are important.

Scale Analysis of the Equations of Motion

- Scale Analysis is a convenient technique for estimating the magnitudes of various terms in the governing equations for a particular type of motion.
- In scale analysis, typical expected values of the following quantities are specified:
 - (a) magnitudes of the field variables ✓
 - (b) amplitudes of fluctuations in the field variables ✓
 - (c) characteristic length, depth and time scales on which these fluctuations occur
- Scale analysis retains the dominant terms in the governing equations and thus simplifies them, and also eliminates the unwanted type of motion.
- Hence it is useful in modeling various types of motion systems.

Consider the case of a “mid-latitude synoptic system” which has the following characteristic scales:

Variable	Order of magnitude	Description
U	10^1 ms^{-1}	Horizontal velocity Scale (u,v)
W	$1 \text{ cms}^{-1} (10^{-2} \text{ ms}^{-1})$	Vertical velocity Scale
L	10^6 m	Length Scale
H	10^4 m	Depth Scale
$\Delta P/\rho$	$10^3 \text{ m}^2\text{s}^{-2}$	Horizontal Pressure Fluctuation Scale
$f_0 = 2\Omega \sin \Phi$	10^{-4} s^{-1}	Coriolis Parameter
$T = L/U$ ✓	10^5 s	Time Scale
g	10^1 ms^{-2}	Gravity
a	10^7 m	Radius of the earth

Here, we consider only the Horizontal Equations. It can be deduced from horizontal velocity field.

	A	B	C	D	E	F	G
U	$\frac{du}{dt}$	$-2\Omega v \sin \phi + 2\Omega w \cos \phi$	$+\frac{uw}{a}$	$-\frac{uv \tan \phi}{a}$	$= -\frac{1}{\rho} \frac{\partial P}{\partial x}$	$+F_x$	
V	$\frac{dv}{dt}$	$+2\Omega u \sin \phi$		$+\frac{vw}{a}$	$+\frac{u^2 \tan \phi}{a}$	$= -\frac{1}{\rho} \frac{\partial P}{\partial y}$	$+F_y$
Scales	U^2/L	$f_o U$	$f_o W$	UW/a	U^2/a	$\Delta P/\rho L$	$\nu U/H^2$
ms^{-2}	10^{-4}	10^{-3}	10^{-6}	10^{-8}	10^{-5}	10^{-3}	10^{-12}

The molecular friction term is so small that it may be neglected for all motions, except the smallest scale turbulent motions near the ground.

Consider the Vertical momentum Equation and perform the scale analysis.

$$\frac{Dw}{Dt} - \frac{u^2 + v^2}{r} = -\frac{1}{\rho} \frac{\partial p}{\partial z} - g + 2\Omega u \cos \phi + \nu \nabla^2 w$$

$W \cdot U/L$	$U \cdot U/a$	$P_{\text{sfc}}/\rho H$	g	Uf	$\nu W/H^2$
10^{-7}	10^{-5}	10 ✓	10 ✓	10^{-3}	10^{-15}

Largest terms

- ❖ Pressure fluctuations are different for the atmospheric systems such as tornadoes, squall lines, and hurricanes.
- ❖ The horizontal pressure gradient can range over several orders of magnitude for systems of meteorological interest.
- ❖ Therefore, the nature of the dominant terms in the governing equations is crucially dependent on the horizontal scale of the motions.
- ❖ In particular, motions with horizontal scales of only a few kilometers tend to have short time scales so that terms involving the rotation of Earth are negligible, while for larger scales they become very important.
- ❖ Because the character of atmospheric motions depends so strongly on the horizontal scale, this scale provides a convenient method for the classification of motion systems

Total Differentiation

- In order to apply the basic laws in the Eulerian frame, it is necessary to derive a relationship between the rate of change of a field variable following the motion and its rate of change w.r.t. a fixed point. The former is called the **substantial**, the **total**, or the **material** derivative and the latter is called the local derivative (which is merely the partial derivative w.r.t. time).
- Suppose the temperature measured on a balloon which moves with the wind, is 'T' at the point x, y, z and time t. For the air parcel within the balloon, the location (x, y, z) is a function of time 't' so that $x = x(t)$, $y = y(t)$, $z = z(t)$. Following the balloon, T may then be treated as truly a function only of time, and its rate of change is just the total derivative, DT/dt .

Atmospheric Force Balances

Equation of Motion

$$\frac{D\vec{u}}{Dt} = -2\vec{\Omega} \times \vec{U} - \frac{1}{\rho} \vec{\nabla} P + \vec{g} + \vec{F}_r$$

The eastward (u)_Zonal, northward (v)_Meridional and, vertical (w) components of momentum equations respectively:

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Equation of Motion

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$$\frac{dw}{dt} \quad \text{[]} = -\frac{1}{\rho} \frac{\partial P}{\partial z} - g + \text{[]} + F_z$$

	A	B	C	D	E	F	G
U	$\frac{du}{dt}$	$-2\Omega v \sin \phi - 2\Omega w \cos \phi$		$+\frac{uw}{a}$	$-\frac{uv \tan \phi}{a}$	$= -\frac{1}{\rho} \frac{\partial P}{\partial x}$	$+ F_x$
V	$\frac{dv}{dt}$	$+2\Omega u \sin \phi$		$+\frac{vw}{a}$	$+\frac{u^2 \tan \phi}{a}$	$= -\frac{1}{\rho} \frac{\partial P}{\partial y}$	$+ F_y$
Scales	U^2/L	$f_o U$	$f_o W$	UW/a	U^2/a	$\Delta P/\rho L$	$\nu U/H^2$
ms^{-2}	10^{-4}	10^{-3}	10^{-6}	10^{-8}	10^{-5}	10^{-3}	10^{-12}

$$\frac{Dw}{Dt} - \frac{u^2 + v^2}{r} = -\frac{1}{\rho} \frac{\partial p}{\partial z} - g + 2\Omega u \cos \phi + \nu \nabla^2 w$$

$W \cdot U/L$	$U \cdot U/a$	$P_{\text{sfc}}/\rho H$	g	Uf	$\nu W/H^2$
10^{-7}	10^{-5}	10	10	10^{-3}	10^{-15}

Geostrophic motions

Component form:

$$u_g = -\frac{1}{\rho f} \frac{\partial p}{\partial y}$$

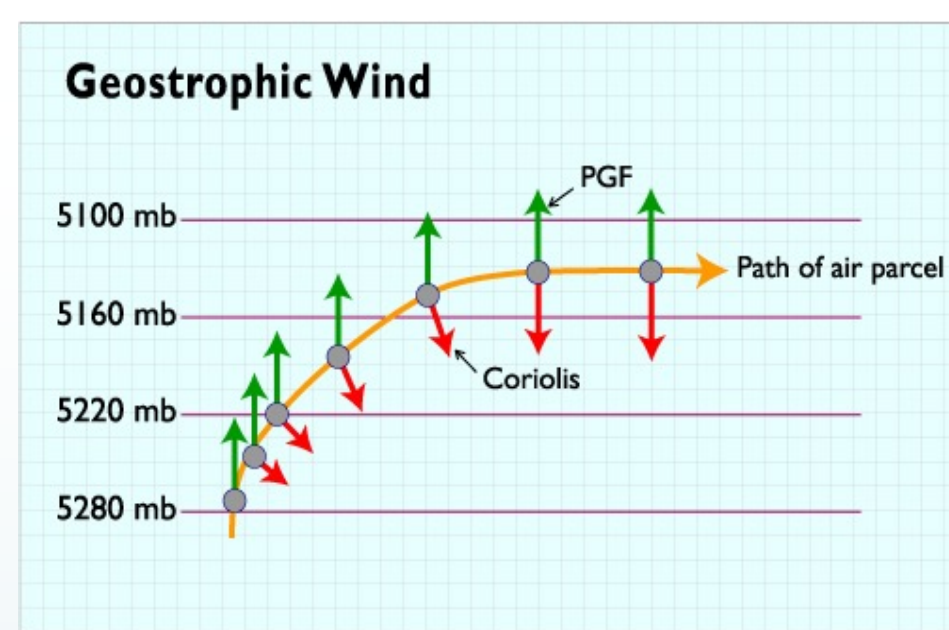
$$v_g = \frac{1}{\rho f} \frac{\partial p}{\partial x}$$

Vector form:

$$\vec{v}_g = \frac{1}{\rho f} \vec{k} \times \nabla p$$

with the Coriolis parameter $f = 2\Omega \sin \phi$.

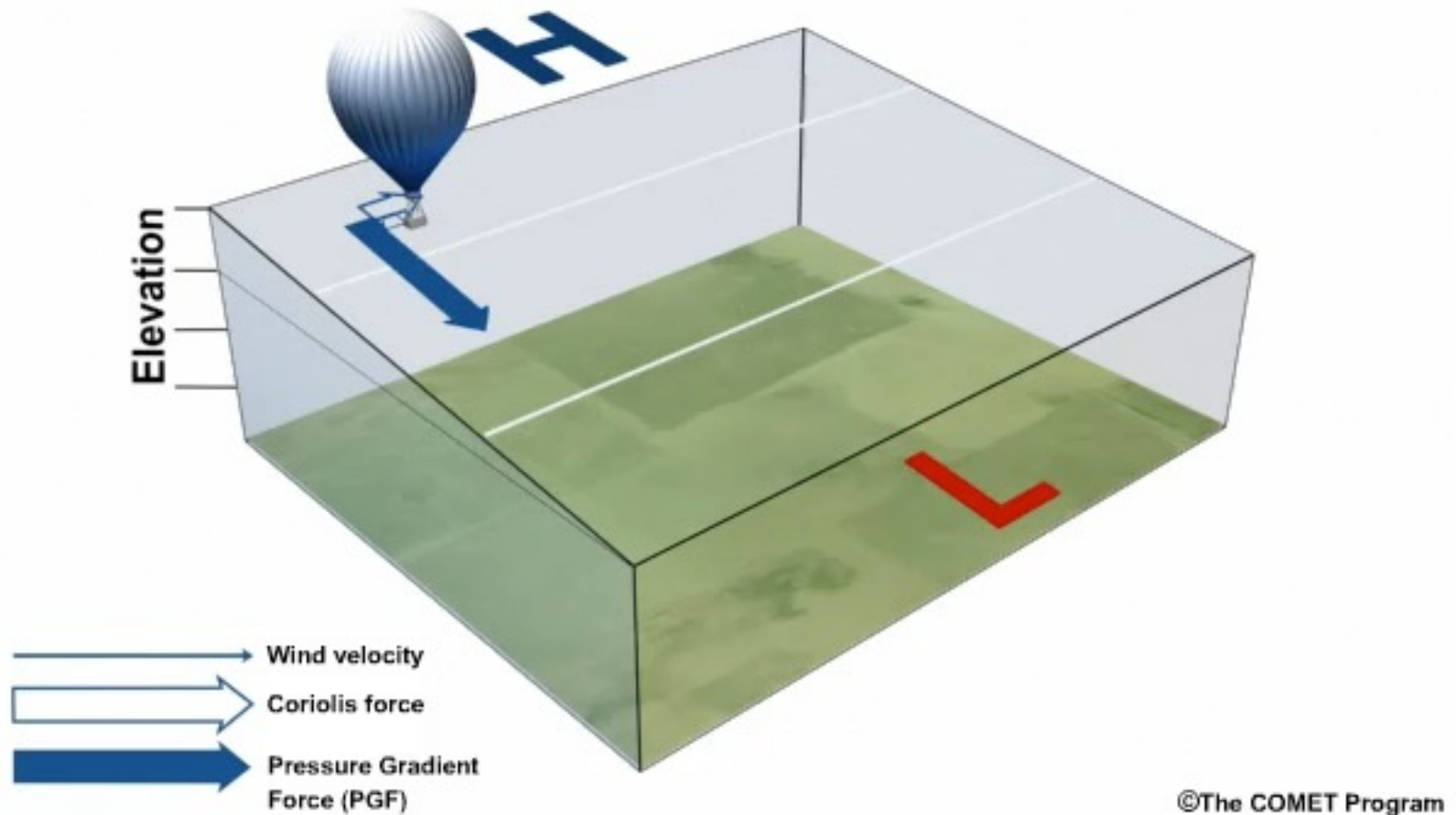
- The geostrophic wind describes the dominant balance between the pressure gradient force and the Coriolis force.
- Just above the top of the boundary layer, the atmosphere is close to geostrophic balance.



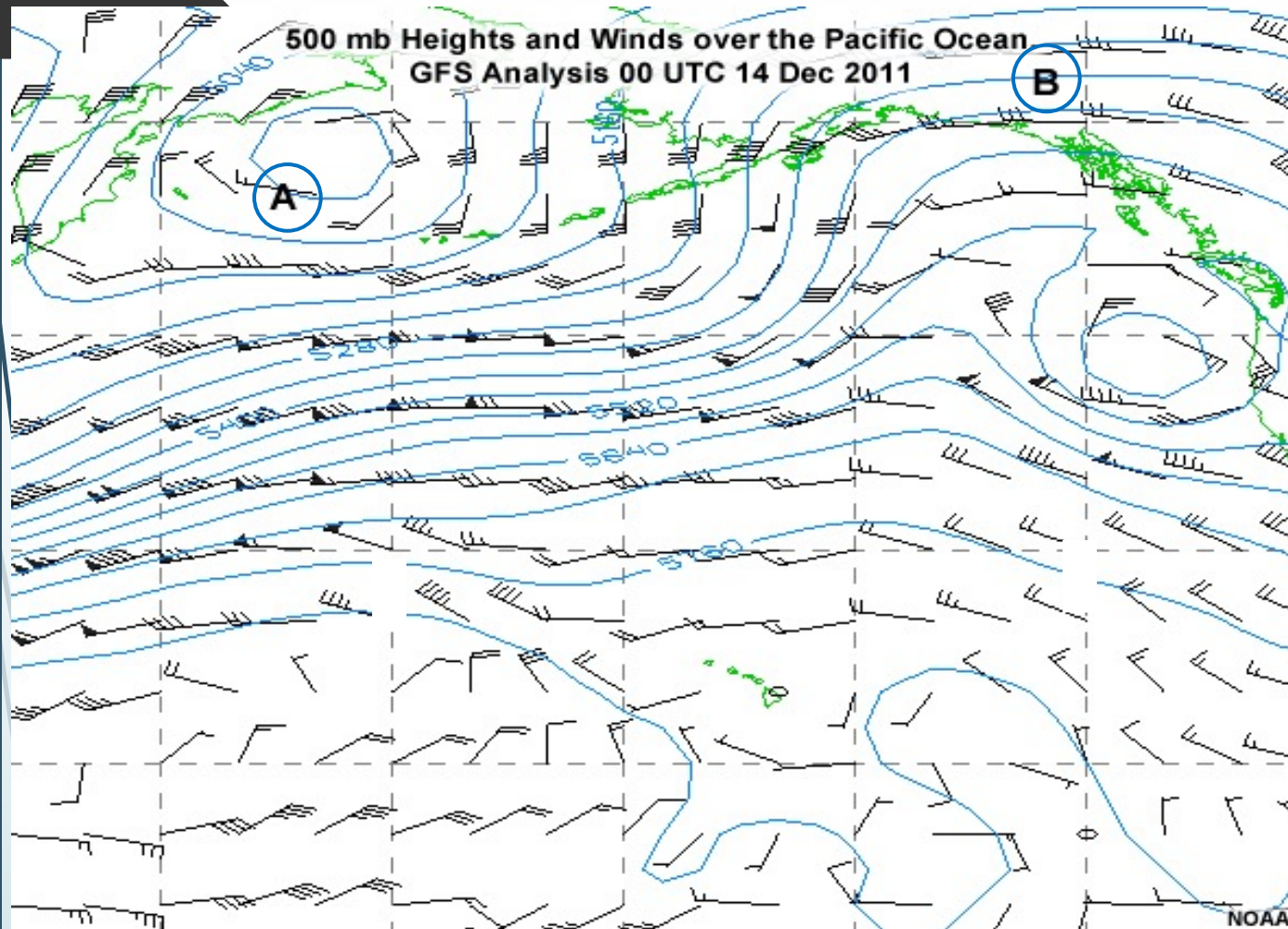
Flow trigger by PGF

Physics of Geostrophic Wind

Movement of Geostrophic wind



Given this chart of 500 mb heights and winds, at which location is the flow most geostrophic? Most non-geostrophic due to centripetal acceleration?

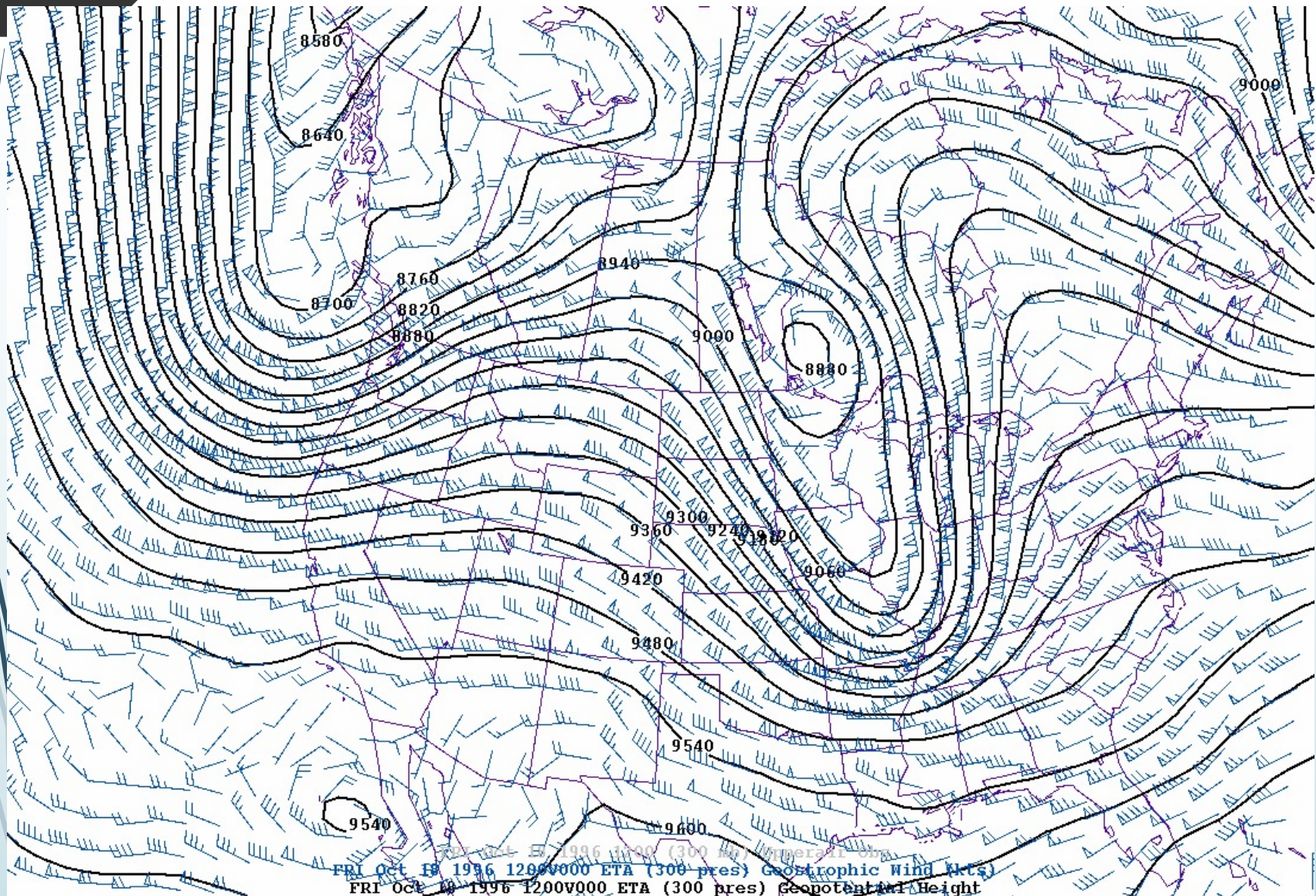


Most geostrophic ?

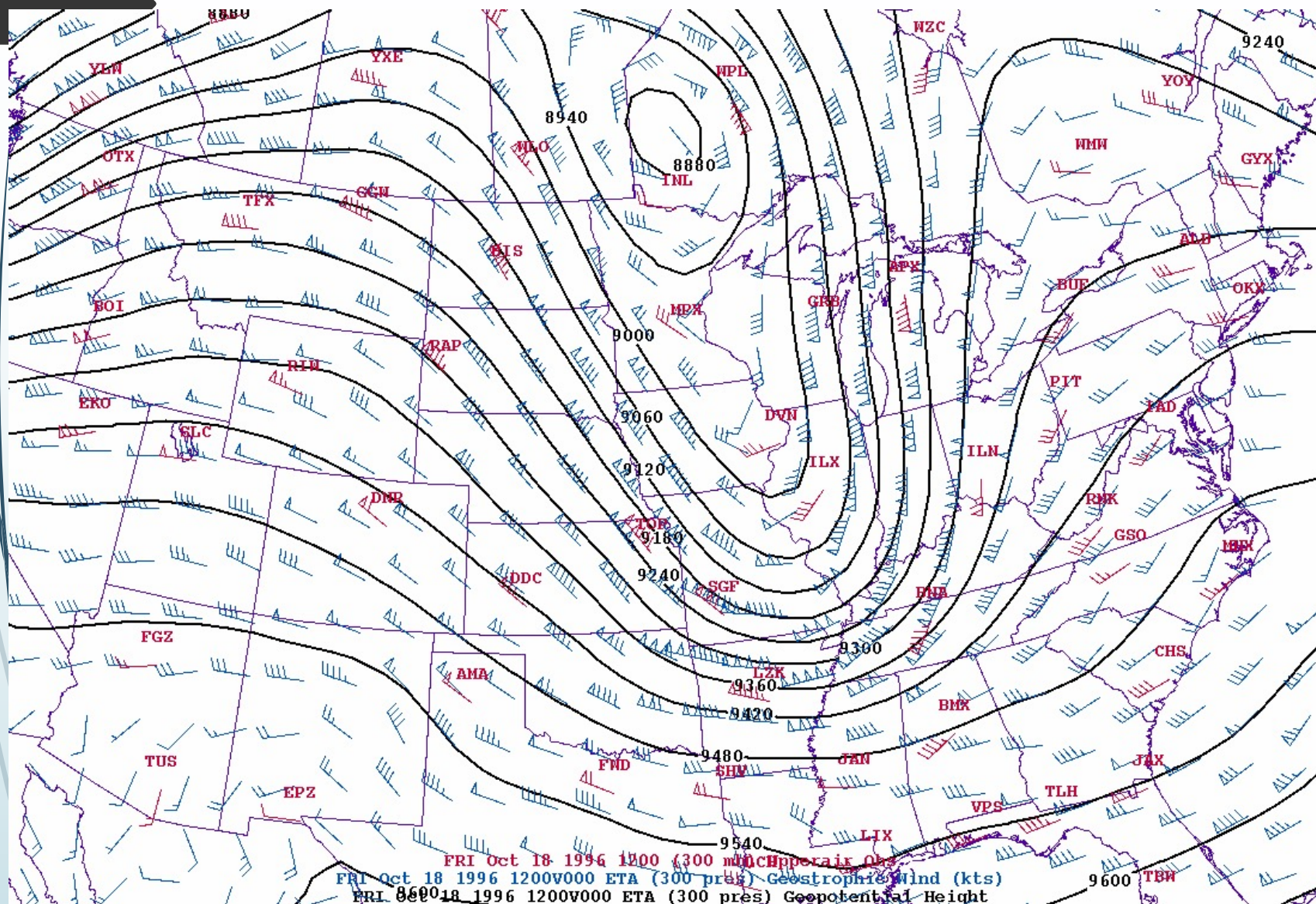
Non geostrophic ?

- The most geostrophic flow tends to occur where contours are straight and evenly spaced.

Geostrophic Wind @ 300 mb (Approx. 9km)



Geostrophic and Observed Wind @ 300 mb (Approx. 9km)



Rossby Number :

A non-dimensional number that is used to estimate the **validity** **Geostrophic approx**/ the importance of rotation

$$\varepsilon = \frac{U}{2\Omega L}$$

$\Omega = 7.3 \times 10^{-5} \text{ rad/sec}$
(Angular speed of Earth's rotation)

- The Rossby number (ε) is the ratio of the characteristic scales for the acceleration and the Coriolis force terms.
- For large-scale flows $\varepsilon \leq 1$

The acceleration term

$$\frac{d\vec{u}}{dt} = O\left(\frac{U^2}{L}\right)$$

The Coriolis force term

$$2\vec{\Omega} \times \vec{u} = O(2\Omega U)$$

Examples: Low and high Rossby number flows

- A typical synoptic scale motion in the atmosphere has a length scale $L \sim 1000$ km and velocity scale $U \sim 20 \text{ ms}^{-1}$.

This is a low Rossby number flow since $\varepsilon < 1$ (eg. The 500 hPa height field)

- In the gulf-stream, $U \sim 100 \text{ cms}^{-1}$ and the characteristic horizontal scale $L \sim 100$ km.

Here $\varepsilon \sim 0.07$ and qualifies as large-scale motion

- Consider a tornado – which has a typical tangential velocity say 30 ms^{-1} at a distance of 300 m from the center of the vortex.

It turns out to be high Rossby number flow ($\varepsilon \sim 700$)

Geostrophic Balance : Important to know

- The direction of the geostrophic wind is **parallel** to the **isobars**.
- The closer the isobars are together, the stronger the magnitude of the geostrophic wind $|v_g|$ (isotachs increase).
- This is very true for large-scale motions away from the equator.
- The geostrophic wind v_g is a good approximation of the real horizontal wind vector, especially over oceans and at upper levels.
- Why?

Hydrostatic Balance

Components of
Equation of Motion

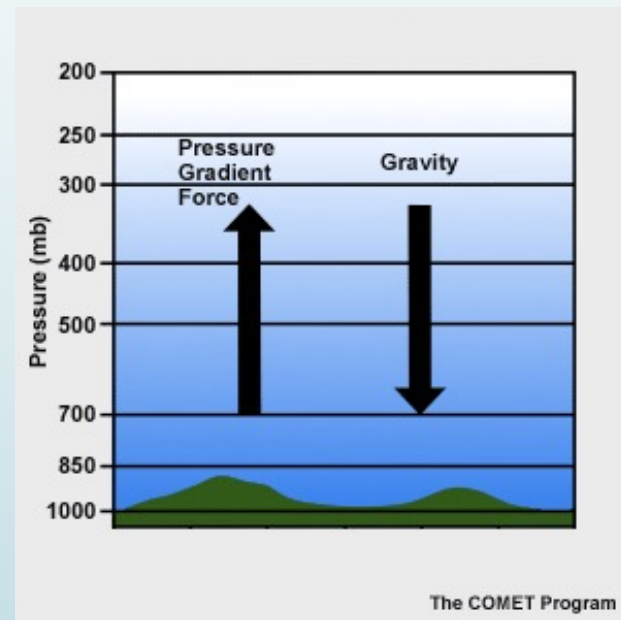
$$\frac{du}{dt} = -\frac{1}{\rho} \frac{\partial P}{\partial x} + 2\Omega v \sin \phi + F_x$$

$$\frac{dv}{dt} = -\frac{1}{\rho} \frac{\partial P}{\partial y} - 2\Omega u \sin \phi + F_y$$

$$\frac{dw}{dt} = -\frac{1}{\rho} \frac{\partial P}{\partial z} - g + \text{[]} + F_z$$

$$\frac{1}{\rho} \frac{\partial p}{\partial z} = -g$$

**Hydrostatic
Balance**



Hydrostatic Balance

Decreasing pressure with height imposes an upward force on any element of fluid (the pressure gradient force), which almost exactly balances the gravitational force due to the mass of atmosphere above.

⇒ the upward pressure gradient force acting on a thin slice of air (p decreases with z) is generally very closely in balance with the downward force due to gravity

The **net upward** force acting on the slice of air

$$\frac{\vec{F}_{PGF}}{A} = \delta p, \quad (\delta p < 0)$$

where A is the horizontal cross section area of the slice.

The **gravitational force** acting on the slice of air

$$\frac{\vec{F}_{gravity}}{A} = \rho g \delta z$$

where g is the acceleration due to gravity (9.8 m s^{-2}),

m is the mass of this thin slice,

ρ is the density of the fluid.

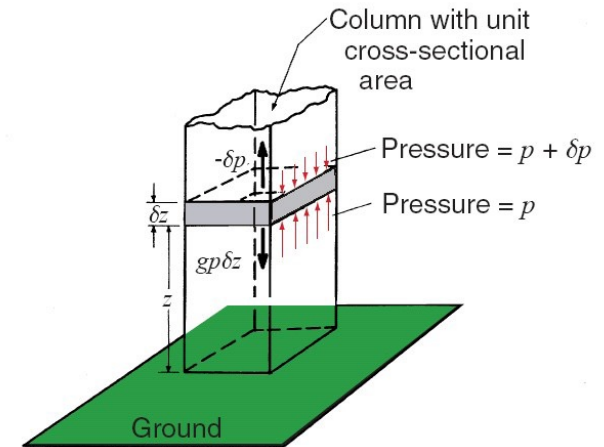
Hydrostatic equilibrium ⇒ $\vec{F}_{PGF} = \vec{F}_{gravity}$

$$\frac{\delta p}{\delta z} = -\rho g$$

$$\lim_{\Delta p, \Delta z \rightarrow 0} \frac{\delta p}{\delta z} = \frac{dp}{dz}$$

$$\Rightarrow \frac{dp}{dz} = -\rho g$$

This balance is referred to as the hydrostatic equilibrium



$$\frac{dp}{dz} \equiv -\rho g$$

Hydrostatic Balance: Important to know

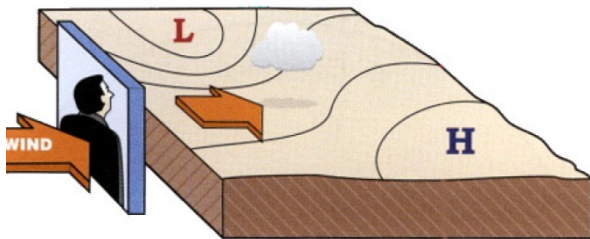
- Hydrostatic approximation can be obtained from the vertical momentum equation by comparing the magnitudes of the different components
- The hydrostatic equation tells us that the rate at which the pressure decreases with height is equal to the air density times the acceleration of gravity (where $-g$ is actually the force of gravity per unit volume). The minus sign indicates that, as the air pressure decreases, the height increases.
- Large-scale atmospheric motions are nearly-hydrostatic
- The air pressure decreases more rapidly with height in cold (more-dense) air than it does in warm (less dense) air.

Atmospheric Balanced Flows

- Balanced flow is often an accurate approximation of the actual flow, and is useful in improving the qualitative understanding and interpretation of atmospheric motion.
- Some of the idealized air motion:

Geostrophic, cyclostrophic, inertial and gradient flows.

Buys-Ballot Law



Buys Ballot's law : In the Northern Hemisphere, if a person stands with his back to the wind, the low pressure area will be on his left (1857)

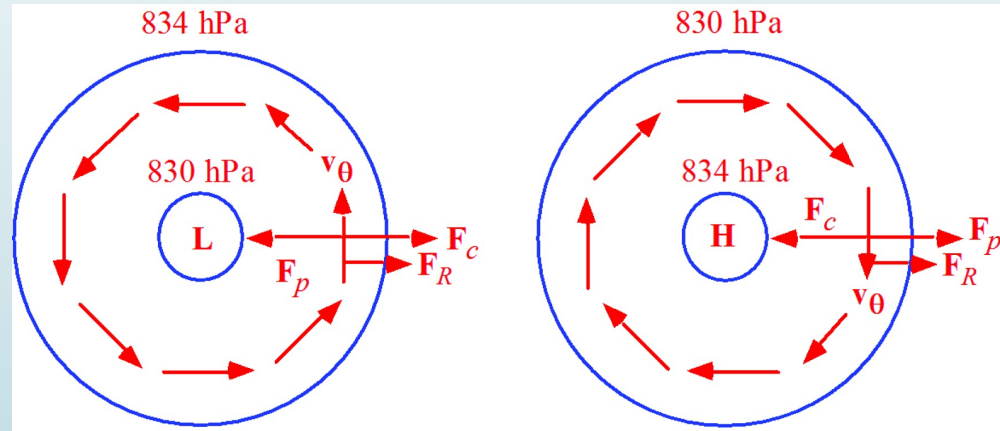
Gradient wind

- The gradient wind is a balance of the Pressure Gradient Force, centrifugal and Coriolis
- Wind which results when the Centrifugal Force, resulting from curved flow, is exactly balanced by the Coriolis and Pressure Gradient Forces.

Momentum Equation in Natural Coordinates:

$$\frac{V^2}{R} + fV = -\frac{\partial\Phi}{\partial n}$$

In a circular flow, the direction of the air motion is constantly changing, so, according to the Newton's second law, there must be a net force acting on air in the direction toward the center of the circle: *centripetal force*. A wind that blows at a constant speed parallel to curved isobars is called the **gradient wind**.




$$V^2 + fRV + R\frac{\partial\Phi}{\partial n} = 0$$

Quadratic Equation

$$V = -\frac{fR}{2} \pm \sqrt{\frac{(fR)^2}{4} - R\frac{\partial\Phi}{\partial n}}$$

The sign of the terms in this three way balance depend on whether the curvature of the flow is cyclonic or anticyclonic.

Cyclostrophic Flow

- ❖ Cyclostrophic flow describes steady flows where **centrifugal force is balanced by pressure gradient force**, and Coriolis force is largely negligible

$$V^2 = -R \frac{\partial \Phi}{\partial n}$$

Cyclostrophic flow arises when V is large or R is small.



Inertial Flow

- ❖ Inertial flow describes steady flows where centrifugal force is balanced by Coriolis force and pressure gradients are largely negligible

$$\frac{V^2}{R} + fV = 0$$

$$V = -fR$$

Item	Name of Wind	Forces	Direction	Magnitude	Where Observed
1	geostrophic	pressure-gradient, Coriolis	parallel to straight isobars with Low pressure to the wind's left*	faster where isobars are closer together. $G = \left \frac{g}{f_c} \cdot \frac{\Delta z}{\Delta d} \right $	aloft in regions where isobars are nearly straight
2	gradient	pressure-gradient, Coriolis, centrifugal	similar to geostrophic wind, but following curved isobars. Clockwise* around Highs, counterclockwise* around Lows.	slower than geostrophic around Lows, faster than geostrophic around Highs	aloft in regions where isobars are curved

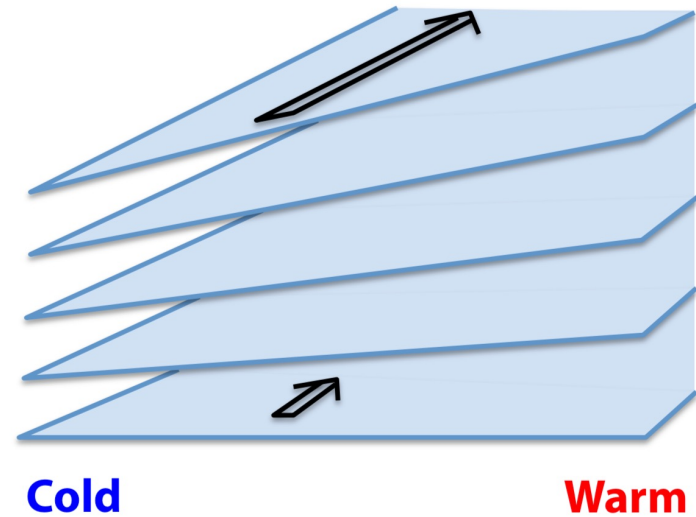
Balanced Motions

5	cyclostrophic	pressure-gradient, centrifugal	either clockwise or counterclockwise around strong vortices of small diameter	stronger for lower pressure in the vortex center	tornadoes, waterspouts (& sometimes in the eye-wall of hurricanes)
6	inertial	Coriolis, centrifugal	anticyclonic circular rotation	coasts at constant speed equal to its initial speed	ocean-surface currents

Thermal Wind

- The thermal wind is a vector difference between the geostrophic wind at an upper level and a lower levels
 - Thickness of layers related to temperature, causes a tilt in the pressure surfaces
 - Change in magnitude of horizontal gradient of pressure then leads to **vertical wind shear**

The thermal wind vector points such that **cold air is to the left** and **warm air is to the right**, parallel to isotherms-”constant temperature lines” (in the northern hemisphere).



A horizontal thermal gradient creates a PGF at upper levels. As you increase in altitude, the pressure gradient between the warm column and the cool column increase. As the PGF increases the magnitude of the wind will increase and so will the Coriolis force (geostrophic winds - looking at an arrow pointing into away from us - into the image).

The vertical change in geostrophic wind is called the geostrophic vertical shear. Since the geostrophic vertical shear is directly proportional to the horizontal temperature gradient, it is also called the Thermal Wind.

Thermal wind is not an actual wind, but a the difference between two winds at different levels.

The Thermal Wind: Vertical shear of the geostrophic wind

geostrophic wind $\Rightarrow$

$$\vec{v}_g = \frac{1}{\rho f} \vec{k} \times \nabla p$$

Recalling that the geostrophic wind is proportional to horizontal pressure-gradient; & making use of the hydrostatic equation & Ideal gas-law, we can derive the “Thermal Wind” equation

$$\frac{\partial \mathbf{V}_g}{\partial \ln p} = -\frac{R}{f} \mathbf{k} \times (\nabla T)_p$$

A relationship for the vertical shear of the geostrophic wind and the horizontal temperature gradient

The thermal wind vector $\mathbf{V}_T$ from pressure level p_0 to level p_1 ($p_1 < p_0$) is given by

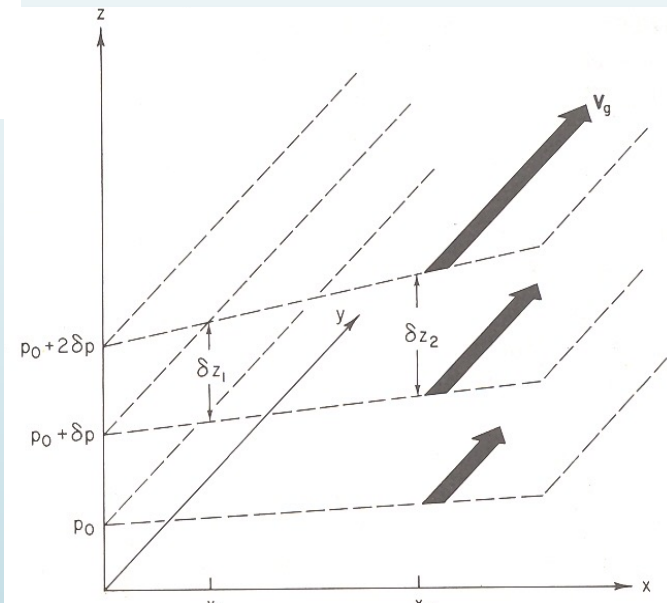
$$\mathbf{V}_T = \mathbf{V}_g(p_1) - \mathbf{V}_g(p_0) = -\frac{R}{f} \int_{p_0}^{p_1} (\mathbf{k} \times \nabla T) d \ln p$$

$\bar{T}$ = Mean temperature in the layer between pressure p_0 and p_1

The x and y components of the thermal wind are given by:

$$u_T = -\frac{R}{f} \left(\frac{\partial \bar{T}}{\partial y} \right)_p \ln \left(\frac{p_0}{p_1} \right)$$

$$v_T = \frac{R}{f} \left(\frac{\partial \bar{T}}{\partial x} \right)_p \ln \left(\frac{p_0}{p_1} \right)$$

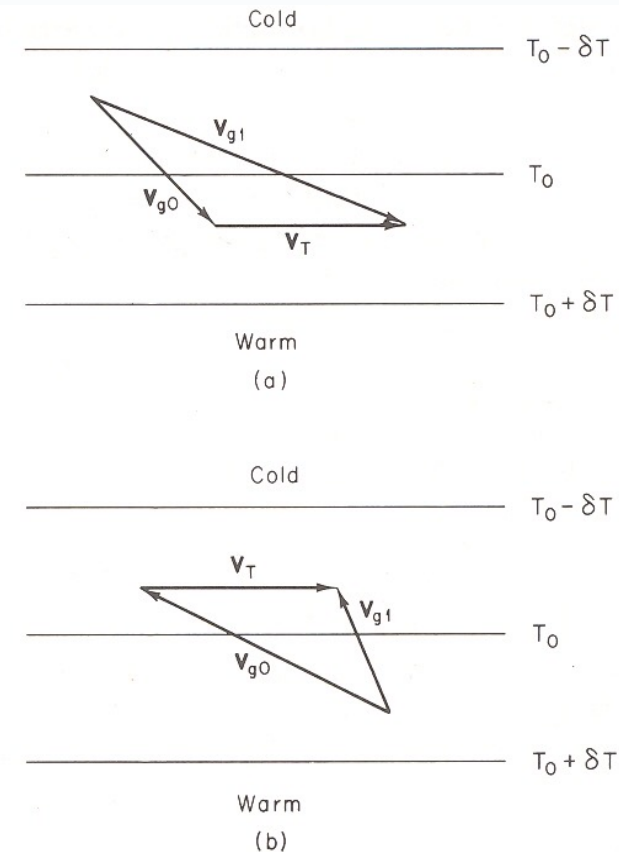


- The thermal wind equation is an extremely useful diagnostic tool for checking analyses of the observed wind and temperature fields for consistency. It can also be used to estimate the mean horizontal temperature advection in a layer

The thermal wind blows parallel to the isotherms (lines of constant temperature / thickness)
 – with the warm air to the right facing downstream in the NH.

Thus, a geostrophic wind which turns **anti-clockwise with height** (backs) is associated with **cold-air advection**

Conversely, with **clockwise turning (veering)** of the geostrophic wind height implies **warm-air advection** in the layer



Remember this thumb rule for thermal wind balance:

Zonal wind shear (X-Z plane) is proportional to meridional (Y-dir) temperature gradient

Meridional wind shear (Y-Z plane) is proportional to zonal (X-dir) temperature gradient